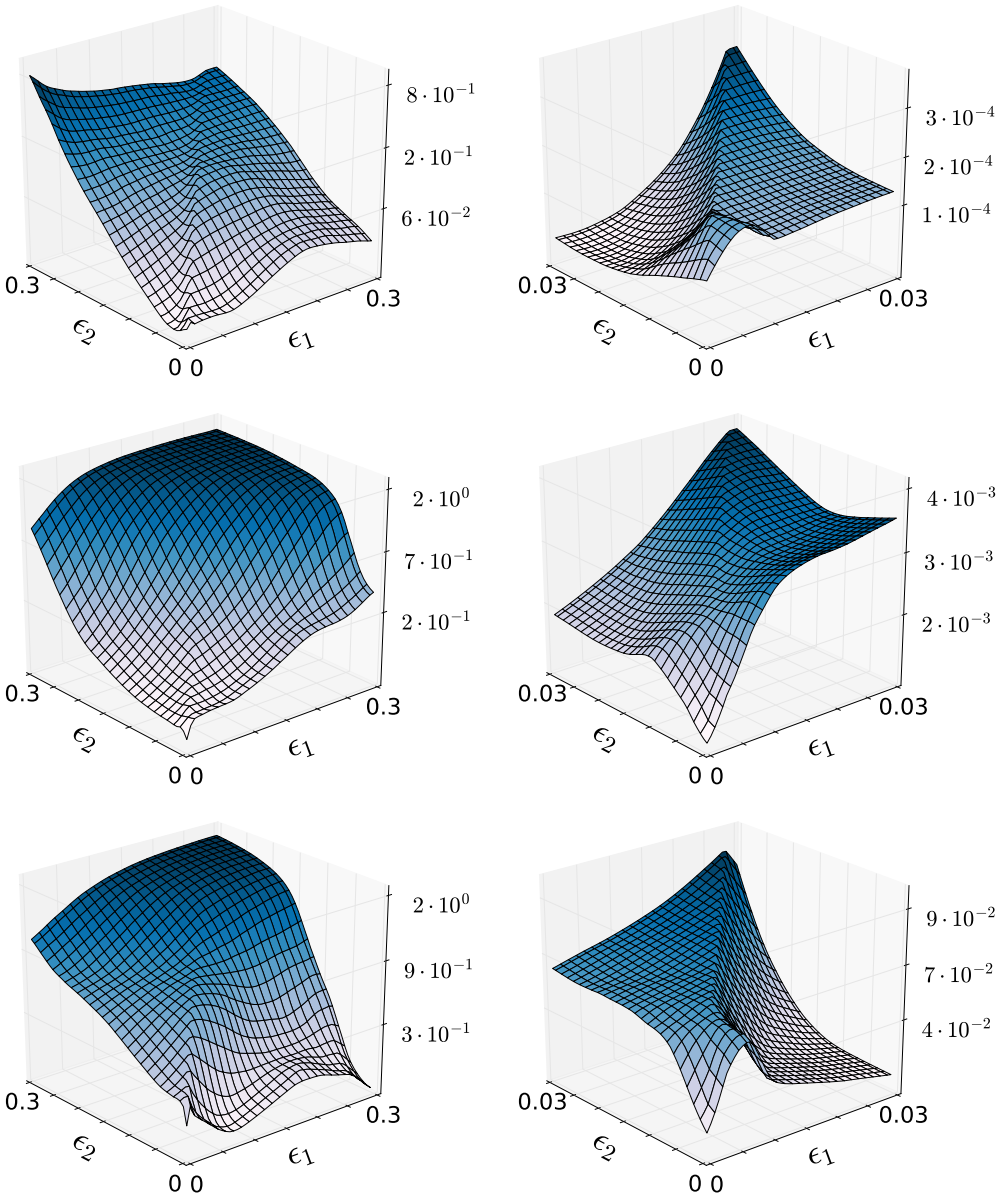

(a) Loss surface for model A_{adv} (log-scale).

(b) Zoom in of the loss for small ϵ_1, ϵ_2.

Figure 5: **Illustrations of the local curvature artifacts introduced by adversarial training on MNIST.** We plot the loss of model A_{adv} on samples of the form $x^* = x + \epsilon_1 \cdot g + \epsilon_2 \cdot g^{\perp}$, where g is the signed gradient of model A_{adv} and $g^{\perp}$ is an orthogonal adversarial direction, obtained from model B. The right-side plots are zoomed in versions of the left-side plots.